

Linfei Li

(+86) 183-8740-0236 · cslinfeili@tongji.edu.cn · lif314.github.io · Google Scholar

EDUCATION

Tongji University

Ph.D. Student in Computer Science, GPA: 90.67/100

Shanghai, China

09/2023 – 03/2029

(5.5-year program)

Advisors: Prof. Lin Zhang and Prof. Ying Shen

Tongji University

B.S. in Software Engineering, GPA: 4.79/5.0

Shanghai, China

09/2019 – 07/2023

Selected coursework: Computer Vision, Computer Graphics, Data Structures and Algorithms, Programming Languages, Operating Systems, Computer Networks, Software Engineering

SELECTED PUBLICATIONS

**Equal contribution.*

1. **Linfei Li**, Ruining Hu, Lin Zhang, Zhong Wang, Fengyi Zhang, Ying Shen, Binqiang Wang, Xin Zhang. *UniRAP: Towards Unified Part-level Physical Affordance Reasoning and Actionable Perception*. In **NeurIPS**, 2026. (CCF-A)
2. **Linfei Li**, Lin Zhang, Ying Shen. *RealVLG-R1: A Large-Scale Real-World Visual-Language Grounding Benchmark for Robotic Perception and Manipulation*. In **CVPR**, 2026. (CCF-A)
3. Yuchuan Ding*, **Linfei Li***, Lin Zhang, Ying Shen. *RaysUp: Ultra-light Universal Feature Upsampling via Geometry-Aware Ray Representation*. In **ECCV**, 2026. (CCF-B)
4. **Linfei Li**, Lin Zhang, Zhong Wang, Ying Shen. *SmartSplat: Feature-Smart Gaussians for Scalable Compression of Ultra-High-Resolution Images*. In **AAAI**, 2026. (CCF-A)
5. **Linfei Li**, Lin Zhang, Zhong Wang, Fengyi Zhang, Zelin Li, Ying Shen. *Representing Sounds as Neural Amplitude Fields: A Benchmark of Coordinate-MLPs and a Fourier Kolmogorov-Arnold Framework*. In **AAAI**, 2025. (CCF-A)
6. **Linfei Li**, Lin Zhang, Zhong Wang, Ying Shen. *GS3LAM: Gaussian Semantic Splatting SLAM*. In **ACM MM**, 2024. (CCF-A)
7. Zhanbo Shi, Lin Zhang, **Linfei Li**, Ying Shen. *Towards Audio-Visual Navigation in Noisy Environments: A Large-Scale Benchmark Dataset and an Architecture Considering Multiple Sound Sources*. In **AAAI**, 2025. (CCF-A)
8. **Linfei Li**, Fengyi Zhang, Zhong Wang, Lin Zhang, Ying Shen. *INR-Bench: A Unified Benchmark for Implicit Neural Representations in Multi-Domain Regression and Reconstruction*. Technical Report, 2025.

RESEARCH PROJECTS

National Key R&D Program - Next-Generation AI

Beijing, China

Sub-project Technical Lead: Multimodal LLM Cross-Modal Trustworthy Verification

12/2025 – Present

- Lead the technical roadmap and core algorithm development for trustworthy cross-modal reasoning in multimodal large language models.

Changchun Science & Technology Breakthrough Program

Changchun, China

Sub-project Technical Lead: Multi-Sensor Fusion-Based AR Navigation

01/2025 – 12/2025

- Led camera-IMU-LiDAR calibration and designed the multi-sensor fusion framework for in-vehicle AR navigation.

Tongji University Scenario Validation Program

Shanghai, China

Sub-project Technical Lead: Multi-Robot Collaborative Localization and Mapping

11/2024 – 11/2025

- Designed the multi-robot SLAM architecture; led pose-graph optimization and dense semantic mapping from simulation through real-robot deployment.

ACADEMIC SERVICE

Reviewer: CVPR 2026; NeurIPS 2026; ECCV 2026; AAAI 2025–2026; ICML 2025; ACM MM 2025

HONORS & AWARDS

- Tongji University Outstanding Doctoral Student Scholarship 2025
- Tongji University Outstanding Doctoral Student 2025
- Outstanding Graduate of Tongji University 2023
- First Prize, National Mathematical Contest in Modeling (Shanghai Division) 2021
- National Endeavor Scholarship 2020, 2021, 2022
- Tongji Scholarship of Excellence 2020, 2021, 2022

TECHNICAL SKILLS

Research: Multimodal large language models, vision-language models, embodied AI and robot manipulation, visuo-tactile perception, reinforcement learning for robotics, dense prediction, 3D reconstruction

Robotics: Franka Emika robot arms, dexterous hands, tactile sensors, humanoid robots, camera-LiDAR-IMU calibration

Programming: Python, C/C++, Java **Frameworks:** PyTorch, JAX, Hugging Face, ROS/ROS2, Isaac Sim, L^AT_EX